

MOCTAR SANKARA

DATA ENGINEER | ETL/ELT PIPELINES | DISTRIBUTED DATA PROCESSING | AI-ASSISTED DATA PLATFORMS

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PROFESSIONAL SUMMARY

Data Engineer with 7 years of professional experience across data engineering, data analytics, product analytics, and financial technology. Experienced in designing, building, and operating production data pipelines and analytical platforms. Owned distributed workflows processing over 100 TB of data and supported data-quality and SLA-monitoring environments containing over 100 million records. Built in end-to-end data platform delivery, dimensional modeling, AI-assisted development, automated validation, metadata management, production troubleshooting, performance optimization, technical documentation, and cross-functional partnership.

TECHNICAL SKILLS

Programming and Query Languages: Python, SQL, PySpark, Bash, LookML

Data Engineering: ETL/ELT, Batch Processing, Distributed Data Processing, Automated Data Pipelines, API Ingestion, Incremental Processing, Schema Evolution

Cloud & Data Platforms: AWS ecosystem, Snowflake, PostgreSQL

Data Modeling & Quality: Dimensional Modeling, Star Schemas, Fact and Dimension Tables, dbt Tests, Source-to-Target Validation, SLA Monitoring, Data Freshness, Root-Cause Analysis

Orchestration & DevOps: Airflow, dbt Core, Jenkins, Git, GitLab CI/CD, Unit and Integration Testing, Poetry

AI & Analytics: Claude Code, Streamlit, AI-Ready Metadata, NL-to-SQL, Looker, Tableau, Power BI, Superset, Metabase

PROFESSIONAL EXPERIENCE

BitSight Technologies Inc. | Boston, MA

Data Engineer, AI and Machine Learning | September 2025 – Present

- Delivered curated data products to internal and external customers through Jenkins-orchestrated ETL workflows processing approximately 100 TB of data using Bash, Amazon EMR, PySpark, and Amazon S3.
- Improved data pipeline development speed by over 35x by building an AI-assisted Airflow DAG generator using Claude Code while preserving testing, code review, and human approval controls.
- Reduced data transformation development time by 20x by building an AI-assisted dbt workflow for data modeling and documentation.
- Developed a production-ready natural-language-to-SQL application using OpenAI LLMs, LangChain, Streamlit, Snowflake, and AWS (EC2, Secrets Manager), achieving 95%+ accuracy across ~200 evaluation questions.
- Orchestrated a metadata and data dictionary platform for ~15 database tables, creating a governed data foundation for AI applications through version control, Python package distribution, and human-in-the-loop validation.
- Improved delivery of monthly analysis and testing 5x faster through development of incremental dbt models for rating-methodology changes.
- Reduced analytical data warehouse maintenance effort 20% by engineering a schema-evolution ETL pipeline that automatically handled schema changes while preserving historical data comparisons using Python and SQL.
- Strengthened data engineering capabilities by mentoring an intern for over 3 months across building scalable data pipelines and production-ready dashboard using Airflow, S3-to-Snowflake architecture, data validation, and Looker.
- Reduced ETL architecture-design time by over 88% by building an AI agent for converting natural-language system descriptions into technical diagrams using Claude, Python, Streamlit, and Mermaid.
- Centralized reusable AI tools in an auto-refreshing GitLab repository, enabling engineers to synchronize company-wide skill libraries from a single source using Gitlab CI/CD functionalities.
- Accelerated Data Science development by building reusable datasets, reducing repetitive preparation of raw production data using PySpark, Python, snowflake and Amazon EMR.

Data Analyst | September 2022–August 2025

- Reduced document-review effort by 90% and generated ~\$165K in annual savings by building an end-to-end LLM workflow for breach information extraction using Python, APIs, OpenAI model, Jenkins, and Amazon S3.
- Delivered reliable daily analytical datasets by building and owning a scalable pipeline transforming tens of millions of data records in Amazon S3 using Python, PySpark, Snowflake, and dbt.
- Developed Python packages for AWS authentication, IAM role assumption, and entity-resolution functionality, improving development time by 5x faster by using Python and GitLab Package Registry through GitLab CI/CD.

- Developed workflow reliability and SLA compliance dashboards tracking execution status, runtime, record volume and completion rates; improving monitoring of data across ~100 million records using Python, dbt and Looker.
- Orchestrated model-monitoring platforms for internet asset attribution metrics, improving visibility into model performance and data-processing reliability.
- Processed more than 80 TB of cyber security data support internet asset attribution analytics, modeling, and reporting using PySpark and Amazon EMR.
- Evaluated cybersecurity health of large Federal Reserve-regulated financial institutions for regulatory and supervisory purposes, enhancing policy maker decision making capabilities.
- Contributed to enhancing Google's Minimum Viable Secure Product framework by conducting SaaS industry security benchmark evaluation through collaboration with Cloud researchers from Google.
- Supported a strategic partnership with Moody's by designing and maintaining cybersecurity governance survey data from more than 2,000 companies in an analytics-ready schema delivered through Amazon S3 and PostgreSQL.
- Resolved Belgium's country rating escalation by analyzing large-scale historical cybersecurity data for identifying the factors behind the rating decline using Amazon EMR, S3 and PySpark.

Pod Foods Co. | San Francisco, CA

Data Analyst, Product | November 2021 – September 2022

- Optimized user experience by analyzing behavior and product performance, driving a 10% increase in customer satisfaction.
- Optimized executive decision-making metrics by 20% by designing cross-functional Tableau, Power BI, and Superset dashboards for financial, operational, and product KPIs.
- Deployed predictive customer-churn and product-recommendation ML models, driving a 15% increase in customer retention for the sales organization.
- Recommended product improvements after conducting more than five user-discovery studies, translating customer behavior, usage patterns, and unmet needs into findings that informed product decisions.
- Partnered with product, engineering, and operations teams to use Snowflake and Python to diagnose and resolve systemic business bottlenecks, improving core product KPIs.

Financial Technology Analyst | October 2019 – October 2021

- Engineered and launched real-time financial analytics solutions for over 250 customers, elevating client platform capability by 15% through cross-functional partnerships.
- Reduced platform operating costs by 10% by applying root-cause analysis to cross-functional business metrics and delivering data-driven action plans.
- Increased operational efficiency by more than 15% by evaluating and improving internal financial processes, systems, and reporting workflows.
- Supported forecasting, budgeting, resource allocation, and periodic reporting by developing complex financial models and presenting recommendations to business leaders.
- Evaluated project feasibility and guided investment decisions by analyzing financial performance, operating costs, resource requirements, and expected business outcomes.
- Supported forecasting, budgeting, resource allocation, and periodic reporting by developing financial models and presenting recommendations to business leaders, improving the clarity of planning and resource decisions.

EDUCATION

Hult International Business School | San Francisco, CA
Master of Science in International Business | 2019

University of Ghana Business School | Accra, Ghana
Master of Business Administration in Finance | 2018

Ecole de Commerce et de Gestion | Abidjan, Côte d'Ivoire
Bachelor of Science in Finance and Accounting | 2016

CERTIFICATIONS

Google Data Analytics Professional Certificate | Google | 2022